The Gram double pencil: a spectral census for the zeros of ζ, and an Euler-anchored regulator

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Abstract

We attach to each window of the critical line a pair of Hankel matrices computed by contour integration of ξ'/ξ, and prove that the number of zeros off the critical line in the window equals the negative inertia of the first matrix (Theorem 2); the remaining spectral data of the pair counts the zeros with multiplicity, their distinct support, and their ordinates (Theorem 5). Both statements are unconditional. The Riemann Hypothesis is thereby equivalent to a Stieltjes-type positivity condition, window by window, in a quotient chart obtained from the functional equation (Proposition 6). We then construct, in the half plane of absolute convergence and with no reference to zeros, an Euler-anchored double pencil which is unconditionally positive semidefinite (Proposition 8), and prove that its spectrum can be transported isospectrally by an explicit finite-dimensional regulator whenever a bank response matrix is nonsingular (Theorem 9). The remaining obligations are stated precisely in §8.

1 Setup

Let ξ be the completed Riemann zeta function and put

$$A(z) := \xi\left(\tfrac{1}{2} - iz\right),$$

an entire function of order one, even ($A(-z) = A(z)$, the functional equation), real on $\mathbb{R}$, whose zeros are $z_\rho = \gamma_\rho + i(\frac{1}{2} - \beta_\rho) \cdot (-1)$ for $\rho = \beta_\rho + i\gamma_\rho$ a nontrivial zero of ζ; in particular

$$\rho \text{ on the critical line} \iff z_\rho \in \mathbb{R}.$$

Nonreal zeros of A occur in conjugate pairs $z, \bar{z}$ with equal multiplicities (conjugate symmetry of ξ on the real axis).

Definition 1 (window moments)**.** Let $W = (a, b)$ with A nonvanishing on $\partial W \times \{|y| \le Y\}$, $Y > \frac{1}{2}$, and let ∂R_W be the positively oriented rectangle boundary. Put

$$\mu_k(W) := \frac{1}{2\pi i} \oint_{\partial R_W} z^k \frac{A'(z)}{A(z)}\, dz, \qquad k \ge 0,$$

and for $n \ge 1$,

$$H_n(W) := \left(\mu_{\alpha+\beta}(W)\right)_{\alpha,\beta=0}^{n-1}, \qquad H_n^{(1)}(W) := \left(\mu_{\alpha+\beta+1}(W)\right)_{\alpha,\beta=0}^{n-1}.$$

By the argument principle, with $x_1, \ldots, x_r$ the distinct real zeros of A in the window (multiplicities $a_i \geq 1$) and $z_j, \bar{z}_j$, $j = 1, \ldots, q$, the distinct conjugate pairs (equal multiplicities $b_j \geq 1$),

$$\mu_k(W) \;=\; \sum_{i=1}^{r} a_i x_i^k \;+\; \sum_{j=1}^{q} b_j \bigl(z_j^k + \bar{z}_j^k\bigr). \tag{1}$$

All $\mu_k(W)$ are real and computable from A alone; no knowledge of the zeros is used. Write $m := r + 2q$ for the number of distinct support points.

2 The inertia identity

Theorem 2 (inertia)**.** *For every $n \geq m$,*

$$\operatorname{inertia} H_n(W) \;=\; \bigl(r + q,\; q,\; n - r - 2q\bigr),$$

where inertia *denotes (positive, negative, zero) eigenvalue counts. In particular*

$$n_-\bigl(H_n(W)\bigr) \;=\; q, \qquad \textit{and} \qquad H_n(W) \succeq 0 \iff q = 0,$$

i.e. the negative inertia counts exactly the off-critical-line conjugate pairs.

Proof. Let $v(t) := (1, t, \ldots, t^{n-1})^T$. By (1),

$$H_n \;=\; \sum_{i=1}^{r} a_i\, v(x_i) v(x_i)^T \;+\; \sum_{j=1}^{q} b_j \Bigl[v(z_j) v(z_j)^T + v(\bar{z}_j) v(\bar{z}_j)^T\Bigr].$$

Write $v(z_j) = A_j + iB_j$ with $A_j, B_j \in \mathbb{R}^n$; then $v(\bar{z}_j) = A_j - iB_j$ and

$$v(z_j) v(z_j)^T + v(\bar{z}_j) v(\bar{z}_j)^T \;=\; 2A_j A_j^T - 2B_j B_j^T, \tag{2}$$

the minus sign being the whole mechanism. Hence $H_n = UJU^T$ with

$$U = \bigl[v(x_1) \cdots v(x_r)\; A_1\, B_1 \cdots A_q\, B_q\bigr], \qquad J = \operatorname{diag}(a_1, \ldots, a_r,\, 2b_1, -2b_1, \ldots, 2b_q, -2b_q),$$

so $\operatorname{inertia} J = (r + q, q, 0)$.

U has full column rank m: the complex matrix $V = [v(x_1), \ldots, v(x_r), v(z_1), v(\bar{z}_1), \ldots]$ is Vandermonde on m distinct nodes with $n \geq m$, so its leading $m \times m$ minor equals $\prod_{\alpha<\beta}(s_\beta - s_\alpha) \neq 0$ and $\operatorname{rank} V = m$; replacing each pair $(v(z_j), v(\bar{z}_j))$ by $(A_j, B_j) = \bigl(\frac{v(z_j)+v(\bar{z}_j)}{2}, \frac{v(z_j)-v(\bar{z}_j)}{2i}\bigr)$ is an invertible change of columns, so $\operatorname{rank} U = m$.

Take a thin QR factorisation $U = QR$ with $Q^T Q = I_m$ and R invertible, and extend Q to an orthogonal $\widetilde{Q} = [Q, Q_\perp]$. Then

$$\widetilde{Q}^T H_n \widetilde{Q} = \begin{pmatrix} RJR^T & 0 \\ 0 & 0_{n-m} \end{pmatrix},$$

and by Sylvester's law of inertia $\operatorname{inertia}(RJR^T) = \operatorname{inertia}(J)$. The zero block contributes $n - m$ null directions. $\square$

Remark 3. The hypothesis $n \geq m$ is necessary: an off-line pair is invisible to a Hankel matrix smaller than the distinct support count. Since $\mu_0(W)$ is the number of zeros in W with multiplicity and $m \leq \mu_0(W)$, the choice $n := \lceil \mu_0(W) \rceil + 1$ is admissible and computable from the bank alone.

Remark 4 (collisions). If a conjugate pair approaches $\mathbb{R}$, the corresponding negative eigenvalue tends to 0^- and at collision becomes a null direction; the invariant through such an event is the inertia, not any individually ordered eigenvalue. Consequently the correct target is non-negativity $H_n \succeq 0$ (equivalently $n_- = 0$), never strict positivity: the $n - m$ structural nulls and the collision nulls are expected.

3 The census, and the equivalent form of RH

Theorem 5 (spectral census)**.** *Let $n \geq m$. From the spectral data of $H_n(W)$ and the value $\mu_0(W)$:*

$$\mu_0 = \textit{number of zeros in } W \textit{ with multiplicity}, \qquad \operatorname{rank} H_n = m = r + 2q,$$

$$n_-(H_n) = q, \qquad n_+(H_n) - n_-(H_n) = r,$$

and $\mu_0 > \sum_i 1 + \sum_j 2$ (i.e. μ_0 exceeds the distinct count) if and only if some zero in W is multiple. Moreover the generalized eigenvalues of the pencil $\left(H_n^{(1)}, H_n\right)$ restricted to $\operatorname{ran} H_n$ *are exactly the support points x_i and $z_j, \bar{z}_j$.*

Proof. The first four items are Theorem 2 together with $\mu_0 = \sum a_i + 2 \sum b_j$. For the pencil statement, $H_n = UJU^T$ and $H_n^{(1)} = UJ\Sigma U^T$ where Σ is the diagonal matrix of support points in the basis of columns of U (Vandermonde shift identity $v(t)^T \cdot t =$ shifted moments); on $\operatorname{ran} H_n$ the pencil is therefore similar to Σ. This is Gaussian quadrature for the measure (1). $\square$

Proposition 6 (quotient chart; Stieltjes form)**.** *Let $w = z^2$ and $\nu_k(W) := \mu_{2k}(W)$. Under $z \mapsto w$: a real zero maps to $w > 0$; a conjugate pair $x \pm iy$ ($x \neq 0$) maps to a conjugate pair in w; a pair on the imaginary axis maps to $w < 0$. Consequently, for n large enough in the sense of Theorem 2,*

$$\textit{all zeros of } A \textit{ in } W \textit{ are real} \iff \left(\nu_{\alpha+\beta}\right) \succeq 0 \quad \textit{and} \quad \left(\nu_{\alpha+\beta+1}\right) \succeq 0,$$

the Stieltjes condition for the measure $\sum_\rho m_\rho \delta_{z_\rho^2}$. Hence RH is equivalent to: for every window of a tiling of $\mathbb{R}$, both Hankel matrices of the even moments are positive semidefinite.

Proof. The functional equation makes A even, so the multiset of zeros is symmetric under $z \mapsto -z$ and descends to the w-chart. Reality and non-negativity of the w-support is exactly the Stieltjes moment condition (Hamburger positivity of both $(\nu_{\alpha+\beta})$ and its shift). The equivalence with the location statement is Theorem 2 applied in the w-chart together with the support map. $\square$

Corollary 7 (sign rigidity)**.** *Let $t \mapsto A_t$ be a continuous deformation with zeros varying continuously and no zero crossing ∂W. Then $H_n(t)$ is continuous, and a conjugate pair is created or annihilated at t_0 if and only if an eigenvalue of $H_n(t)$ crosses 0 at t_0; at simultaneous collisions the invariant is the inertia change. Consequently, if $H_n(t) \succeq 0$ for all t, then $n_-(H_n(t)) \equiv 0$ and no off-line pair is ever born. If moreover $H_n(t_*) \succeq 0$ is known at one parameter t_* (for instance in a numerically verified range), the conclusion propagates along the deformation.*

4 Unconditional statements

The following hold with no hypothesis.

(U1) For every window W and $n \geq m(W)$: $q(W) = n_-\big(H_n(W)\big)$; the off-critical-line pair count of a window is the negative inertia of a matrix computed by contour integration of A'/A.

(U2) The census of Theorem 5 holds as stated.

(U3) RH is equivalent to the Stieltjes condition of Proposition 6 on every window of a tiling, equivalently (Corollary 7) to the absence of a negative-sector eigenvalue crossing under the registered deformation.

(U4) For every window contained in the numerically verified range ($|t| \leq 3 \cdot 10^{12}$ by the computations of Platt and Trudgian), $q(W) = 0$ and hence $H_n(W) \succeq 0$: the positivity hypothesis is an established fact on an initial segment of the tiling.

(U5) With $Y > \frac{1}{2}$ the top edge of ∂R_W lies in the half plane of absolute convergence and the bottom edge maps there by the functional equation, giving a decomposition $H_n = H^{\mathrm{pr}} + H^{\mathrm{arch}} + H^{\mathrm{side}}$ into blocks computable from Dirichlet data, Γ-factors and the lateral edges respectively; in a tiling, adjacent lateral contributions cancel pairwise.

5 The Euler-anchored pencil and its regulator

Nothing in this section refers to zeros; the dependency order is

$$\text{Euler/FE bank} \;\to\; (G_1, G_0) \;\to\; \text{real spectrum} \;\to\; \text{contour transport} \;\to\; \text{residues} \;\to\; \text{zeros},$$

and no step may be inverted.

Proposition 8 (Euler anchor)**.** *Fix $s_0 > 1$ and $N \geq 1$, and set*

$$G_\ell[j,k] \;:=\; \sum_{n\geq 2} \Lambda(n)\, n^{-s_0}\, (\log n)^{\,j+k+\ell}, \qquad 0 \leq j,k < N, \quad \ell \in \{0,1\}.$$

The series converge absolutely, and G_0, G_1 are the Hankel matrices of the positive measure $\sum_n \Lambda(n) n^{-s_0} \delta_{\log n}$ supported in $[\log 2, \infty)$. Hence unconditionally

$$G_0 \succeq 0, \qquad G_1 \succeq 0,$$

and $\operatorname{spec}(G_1, G_0) \subset [\log 2, \infty)$ consists of the Gauss quadrature nodes of that measure.

Proof. Absolute convergence for $s_0 > 1$ is standard. Positivity is the Hamburger/Stieltjes criterion for a positive measure on $[\log 2, \infty)$: for $c \in \mathbb{R}^N$ and $P(x) = \sum_j c_j x^j$, $c^T G_0 c = \int P^2\, d\sigma \geq 0$ and $c^T G_1 c = \int x P^2\, d\sigma \geq 0$. $\square$

Theorem 9 (unconditional reverb regulator)**.** *Let $t \mapsto (G_0(t), G_1(t))$ be a smooth family with $G_0(t) > 0$ on the active quotient, and set $S(t) := G_0^{-1/2} G_1 G_0^{-1/2}$, so $S = S^*$ and* $\operatorname{spec} S =$

spec(G_1, G_0). *Suppose the spectrum of $S(t_0)$ is simple, with spectral projections P_a and eigenvectors v_a, and let the transport be controlled,*

$$\dot{S} \;=\; F(u) \;=\; F_0 + \sum_{r=1}^{m} u_r F_r, \qquad F_r = F_r^*.$$

Define $M_{ar} := \langle v_a, F_r v_a \rangle$ and $b_a := -\langle v_a, F_0 v_a \rangle$. If $\det M \neq 0$, then $u := M^{-1} b$ is the unique control with $\Delta\big(F(u)\big) := \sum_a P_a F(u) P_a = 0$, and with

$$K := \sum_{a \neq b} \frac{P_a F(u) P_b}{\lambda_a - \lambda_b}, \qquad K^* = -K,$$

one has $\dot{S} = [S, K]$. Consequently $S(t) = U(t) S(t_0) U(t)^*$ with $\dot{U} = -KU$, U unitary, and

$$\operatorname{spec} S(t) \;=\; \operatorname{spec} S(t_0) \quad \textit{exactly.}$$

In particular the transported pencil remains Hermitian-definite and its generalized spectrum remains real.

Proof. First-order perturbation theory gives $\dot{\lambda}_a = \langle v_a, F v_a \rangle$, so $\Delta(F)$ is precisely the component of the flow able to move eigenvalues; the linear system $Mu = b$ expresses $\Delta(F(u)) = 0$ and is uniquely solvable iff $\det M \neq 0$. For such u, using $\sum_a P_a = I$,

$$[S, K] = \sum_{a \neq b} (\lambda_a - \lambda_b) \frac{P_a F(u) P_b}{\lambda_a - \lambda_b} = \sum_{a \neq b} P_a F(u) P_b = F(u) - \Delta(F(u)) = F(u) = \dot{S}.$$

$K^* = -K$ since $F(u)^* = F(u)$ and the denominators are real and antisymmetric in (a, b). The solution of $\dot{U} = -KU$, $U(t_0) = I$, is unitary, and $\frac{d}{dt}\big(U^* S U\big) = U^*\big(\dot{S} - [S, K]\big) U = 0$. □

Remark 10 (numerical verification). For the Euler anchor of Proposition 8 with $s_0 = 1.5$, $N = 4$, control channels given by reweighting the prime channels $p \in \{2, 3, 5, 7\}$ and raw flow $F_0 = \partial_{s_0} S$: the spectrum is simple $(0.9619, 2.7211, 5.6399, 8.8009)$, $\det M = -0.128$ with $\operatorname{rank} M = 4$ and condition number $3.4 \cdot 10^2$; the solved control annihilates all diagonal drifts to 10^{-12}, and $\|[S, K] - (F - \Delta(F))\| = 1.3 \cdot 10^{-11}$ with $\|K + K^T\| = 1.7 \cdot 10^{-13}$.

Lemma 11 (block regulator through collisions)**.** *Let $S = S^*$ have distinct eigenvalues Λ_α with spectral projections P_α of arbitrary finite multiplicities, and let the flow be controlled as in Theorem 9, $\dot{S} = F(u) = F_0 + \sum_r u_r F_r$, $F_r = F_r^*$. With*

$$\Delta(F) := \sum_{\alpha} P_\alpha F P_\alpha, \qquad K := \sum_{\alpha \neq \beta} \frac{P_\alpha F P_\beta}{\Lambda_\alpha - \Lambda_\beta},$$

one has $[S, K] = F - \Delta(F)$ and $K^* = -K$. To first order the group Λ_α splits by the spectrum of the compression $P_\alpha F P_\alpha$; hence the regulated condition is the block equation $P_\alpha F(u) P_\alpha = 0$ for all α, a linear system in u generalizing $Mu = b$, and when it is solvable the flow $\dot{S} = [S, K]$ is isospectral with multiplicities. Between collisions Theorem 9 applies; across a collision the invariant is the inertia (Corollary 7); the regulator therefore extends piecewise along any transport with finitely many collision events.

Proof. $SP_\alpha = \Lambda_\alpha P_\alpha$ and $\sum_\alpha P_\alpha = I$ give

$$[S, K] = \sum_{\alpha \neq \beta} (\Lambda_\alpha - \Lambda_\beta) \frac{P_\alpha F P_\beta}{\Lambda_\alpha - \Lambda_\beta} = F - \Delta(F),$$

and $K^* = -K$ since $F^* = F$ and the real denominators are antisymmetric in (α, β). The first-order motion of a degenerate group is degenerate perturbation theory: the group splits by the eigenvalues of $P_\alpha F P_\alpha$, which vanish identically iff the block equation holds; the condition is linear in u, and on solution intervals the Lax argument of Theorem 9 is unchanged. $\square$

6 Warp covariance: the bank-generated readout

Theorem 9 fixes the abstract spectrum $\{\lambda_a\}$ of the regulated flow; the same transport, unregulated, moves it. The first chart makes reality manifest, the second carries the physical ordinates, and the map between them — the warp — is not a hypothesis to be assumed but an object the bank constructs. Throughout this section $t \mapsto (G_0(t), G_1(t))$ is a smooth transport of the continued jet pair, whose inputs are values and derivatives of the continued logarithmic derivative along the path — function data; no zero location enters — and the parameter interval is *regular*: $G_0(t) > 0$ on the active quotient and $\operatorname{spec} S_t$ simple, where $S_t := G_0^{-1/2} G_1 G_0^{-1/2}$. The finitely many failures of regularity are the collision and rank-drop events, crossed piecewise by Lemma 11.

Definition 12 (bank velocity field; warp)**.** Let $\lambda_1(t) < \cdots < \lambda_n(t)$ be the eigenvalues of S_t, realized as the generalized eigenpairs $G_1 w_a = \lambda_a G_0 w_a$ normalized by $w_a^T G_0 w_a = 1$, and set

$$d_a(t) := w_a^T \big(\dot{G}_1 - \lambda_a \dot{G}_0\big) w_a \qquad \big(= \langle v_a, \dot{S}_t\, v_a\rangle,\ v_a = G_0^{1/2} w_a\big).$$

Let V_t be the unique real polynomial of degree $\leq n-1$ with $V_t(\lambda_a(t)) = d_a(t)$, and let the *warp* φ_t be the flow of the scalar equation $\dot{x} = V_t(x)$, $\varphi_0 = \mathrm{id}$.

Theorem 13 (warp existence and exact covariance)**.** *On every regular interval the warp is defined on a neighborhood of* $[\lambda_1(0), \lambda_n(0)]$ *and satisfies:*

- *(i)* ***bank determination:*** *V_t, hence φ_t, is computed from $(G_0, G_1)(t)$ and $(\dot{G}_0, \dot{G}_1)(t)$ alone;*
- *(ii)* ***reality and order:*** *$\varphi_t(\mathbb{R} \cap \mathrm{dom}) \subseteq \mathbb{R}$ and $x \mapsto \varphi_t(x)$ is strictly increasing, in particular injective;*
- *(iii)* ***exact covariance:*** *$\varphi_t(\lambda_a(0)) = \lambda_a(t)$ for every a and every t in the interval.*

Proof. (i) is the construction. For (ii): V_t has real coefficients — real nodes, real diagonal drifts of a Hermitian family — so the flow preserves $\mathbb{R}$; the field is polynomial in x with coefficients continuous in t, hence locally Lipschitz in x uniformly on compact t-intervals, so distinct scalar characteristics never cross and φ_t is strictly increasing where defined. For (iii): first-order perturbation theory for the Hermitian-definite pencil gives $\dot{\lambda}_a = w_a^T(\dot{G}_1 - \lambda_a \dot{G}_0) w_a = d_a(t) = V_t(\lambda_a(t))$, so the eigenvalue path $t \mapsto \lambda_a(t)$ solves the initial-value problem defining $\varphi_t(\lambda_a(0))$, and uniqueness makes them equal. Existence: the node characteristics are the globally defined eigenvalue paths; a point between consecutive nodes is trapped between two of them by (ii); and continuous dependence on initial conditions extends the flow to a neighborhood of the compact hull for the same times. $\square$

Theorem 14 (determinant pullback). *Let $p_t(x) := \det(S_t - xI)$, and let I be an open real interval containing the time-0 active spectrum on which the warp is defined. Then on $\varphi_t(I)$,*

$$p_t \;=\; J_t \cdot \big(p_0 \circ \varphi_t^{-1}\big),$$

with J_t continuous and nonvanishing; consequently

$$Z(p_t) \cap \varphi_t(I) \;=\; \varphi_t\big(\operatorname{spec} S_0\big) \;\subset\; \mathbb{R}.$$

Proof. The field is polynomial in x, so φ_t is a strictly increasing C^1 diffeomorphism of I onto $\varphi_t(I)$. On $\varphi_t(I)$ the functions p_t and $p_0 \circ \varphi_t^{-1}$ have the same zero set $\{\varphi_t(\lambda_a(0))\}_a = \{\lambda_a(t)\}_a$ (covariance and injectivity, Theorem 13), each zero simple (simple spectrum on a regular interval). The ratio $J_t := p_t/(p_0 \circ \varphi_t^{-1})$ is continuous and nonvanishing off the common zeros and extends across each with the nonzero limit $p_t'(\lambda_a(t))/\big(p_0 \circ \varphi_t^{-1}\big)'(\lambda_a(t))$. $\square$

Remark 15 (the two architectures, reconciled). The regulated and unregulated flows describe one transport in two charts: labels frozen with a moving readout, or a moving spectrum read through the identity; by Theorem 13 the whole discrepancy is φ_t. This corrects the former statement of obligation (N), which asked for a spectral equivalence $\det(G_1 - \lambda G_0) = C \det(H_n^{(1)} - \lambda H_n)$ with one spectral variable λ and a constant C: under the regulated flow that identity is unsatisfiable — the transported spectrum stays at the anchor nodes while the window spectrum sits at the ordinates, disjoint multisets — and it is replaced by the pullback identity of Theorem 14, in which the change of spectral variable is explicit and bank-generated.

Remark 16 (numerical verification). On the convergent segment (s_0: $1.5 \to 1.2$, $N = 4$, exact jets of $-\zeta'/\zeta$; `tmp/att201_warp_probe.py`): the perturbation drift matches finite differences of the eigenvalue paths to $8.9 \cdot 10^{-8}$; the flow of the interpolated field carries the anchor nodes (1.1916, 4.0395, 9.6214, 19.3384) onto the terminal eigenvalues (2.1714, 9.2832, 23.2371, 47.5293) with maximal error $1.9 \cdot 10^{-10}$; an interior test point remains strictly interior. The exact-jet anchor nodes differ from those quoted after Theorem 9, which realize the truncated prime sum at finite cutoff; the two banks are distinct legitimate realizations, and the drift between them is the truncation observation of the ledger.

7 The conditional theorem

Hypothesis 17 (PSD). For every window W of a tiling of $\mathbb{R}$ and $n = \lceil \mu_0(W) \rceil + 1$, $H_n(W) \succeq 0$; equivalently (Proposition 6) both Hankel matrices of the even moments are positive semidefinite in the quotient chart.

Theorem 18. *Assume Hypothesis 17. Then*

1. *every nontrivial zero of ζ lies on the critical line;*
2. *for each window the spectral data of $H_n(W)$ returns the full census of Theorem 5, with $n_- \equiv 0$;*
3. *the zeros are the real generalized eigenvalues of the explicitly computed Hermitian pencils $\big(H_n^{(1)}(W), H_n(W)\big)$.*

Proof. By Theorem 2, $H_n(W) \succeq 0$ with $n \geq m(W)$ forces $q(W) = 0$: no conjugate pair of zeros of A lies in W. Nonreal zeros occur in conjugate pairs with equal real part, hence both members lie in the same window of the tiling (for a pair on a boundary, apply the argument to a shifted tiling). Every nontrivial zero of ζ corresponds to a zero of A lying in some window, and only finitely many per window by Riemann–von Mangoldt. Hence all zeros of A are real, i.e. all nontrivial zeros of ζ have real part $\frac{1}{2}$. Items (2) and (3) are Theorem 5, whose pencil is Hermitian-definite on $\operatorname{ran} H_n$ under Hypothesis 17. □

8 Remaining obligations

Theorem 18 is conditional on Hypothesis 17. By the rigidity of the moment problem, any pairing whose Gram matrix reproduces the moments $\mu_{j+k}(W)$ *is* the counting pairing of (1), so its positivity is equivalent to the conclusion; a proof of Hypothesis 17 must therefore proceed from the bank side, and the following are the precise steps.

(T) **Contour transport.** Continue the matrix-entry identities (not a claimed positive sum) from the Euler-anchored pair of Proposition 8 to the window pair of Definition 1:

$$G_\ell[j,k] \;=\; \frac{1}{2\pi i}\oint_\Gamma z^{\,j+k+\ell}\frac{A'(z)}{A(z)}\,dz \;+\; D_\ell[j,k],$$

with D_ℓ the explicit pole, Γ-factor and lateral contributions of (U5).

(N) **Pencil neutrality.** Show that the transport can be regulated so that D_ℓ does not move the spectrum. By Theorem 9 it suffices that the bank response matrix M be nonsingular along the transport, which is a finite-dimensional, bank-computable condition; it holds at the Euler anchor. It then suffices to obtain the *spectral* equivalence $\det(G_1 - \lambda G_0) = C\det(H_n^{(1)} - \lambda H_n)$, $C \neq 0$, rather than entrywise equality.

(W) **Warp covariance.** As stated in Hypothesis **??**: the control $u = M^{-1}b$ must be an admissible deformation of the registered bank, and the readout family ϕ_t must satisfy (W1) bank determination, (W2) reality and injectivity, and (W3) terminal identification with the zero ordinates. The invariant of the transport is the label set $\{\lambda_a\}$; the physical ordinates are $\phi_t(\lambda_a)$, and it is the readout, not the flow, that must be shown to land on Z_W.

(C) **Collisions.** At spectral collisions replace the scalar conditions by $P_aF(u)P_a = 0$ on the degenerate eigenspace, treating the deformation piecewise between collisions; these are exactly the rank-drop events of Corollary 7.

Granted (T), (N), (W), (C), the chain runs forward: the Euler-anchored pencil is positive semidefinite unconditionally (Proposition 8); its spectrum is transported isospectrally and remains real (Theorem 9); spectral equivalence identifies it with the window pencil; Theorem 5 identifies the latter's spectrum with the zeros; Theorem 2 converts positivity into $q \equiv 0$; and Theorem 18 concludes.